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U-Pb zircon ages for Late Precambrian igneous rocks in southern Britain

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Abstract: Seven new U-Pb zircon ages for volcanic and intrusive igneous rocks from southern Britain are as follows: 613 ± 4 Ma for the Coedana Granite, Anglesey, and 614 ± 2 Ma for the Padarn Tuff Formation of the Arfon Group, North Wales; 677 ± 2 Ma for a deformed granite of the Malvernian Complex and 566 ± 2 Ma for a rhyolitic tuff in the Warren House Formation, Malvern Hills; 603 ± 2 Ma for a granophyric diorite ('markfieldite') intrusive into the Caldecote Volcanic Formation at Nuneaton; 566 ± 2 Ma for the Uriconian volcanic group, Shropshire, and 560 ± 1 Ma for the Ercall Granophyre, which is intrusive into Uriconian volcanic rocks. Three of the four dated intrusive rocks contain a component of inherited zircon, but the dated volcanic rocks are composed of simple, single-stage zircon populations with no detectable amount of secondary Pb-loss. The ages clarify the timing of Precambrian magmatic events in southern Britain, and conform well with those in neighbouring Avalonian terranes. The new age of the Ercall Granophyre provides a reliable maximum age for the Early Cambrian (Atdabanian–late Tommotian) strata of England, and reconfirms a Precambrian age for the Padarn Tuff Formation and granophyric diorite ('markfieldite') which demonstrably underlie Early Cambrian strata.

The late Precambrian rocks of southern Britain consist of several intrusive igneous suites and lithologically and geochemically distinct volcanic and sedimentary sequences (Dunning 1975; Thorpe 1982). Separation of these suites and sequences by major fault zones (Fig. 1) and generally poor exposure have prohibited precise correlation of the Precambrian rocks in Britain. The most recent review of isotope, palaeomagnetic, and geochemical data for the Precambrian rocks in southern Britain (Thorpe *et al.* 1984) suggests that the cratonic basement formed by the aggregation of juvenile island arcs and associated sedimentary prisms in the time interval from *c.* 900 Ma to *c.* 400 Ma.

The timing of events in this model can be verified by precise and reliable isotopic ages for key Precambrian and lower Palaeozoic rocks in southern Britain. Published isotopic ages for many of these rocks rarely have precisions better than 5%, and some of the Rb-Sr and K-Ar ages are of uncertain reliability because of possible mineralogical alteration and isotopic disturbance in early Palaeozoic time, when the region was involved in an Ordovician phase of arc-magmatism and end-Caledonian (Acadian) collisional orogeny. The new U-Pb zircon ages reported here increase considerably the number of high-precision ages in southern Britain and, together with regional geological relationships, provide a sound basis for comparing the Precambrian geological history of the area with Avalonian terranes in North America and North Africa.

Analytical methods

The procedures for zircon dissolution, separation of U and Pb, and isotope dilution mass spectrometry employing a ^{205}Pb-^{235}U enriched tracer-solution follow those of Krogh (1973, 1982) with modifications detailed in Tucker *et al.* (1991). During this study, total procedure blanks for Pb ranged between 2 and 14 pg, and those for U averaged <1 pg. Isotope ratios were measured by single collector peak-jumping using a VG354 solid-source mass spectrometer equipped with a (Daly-type) secondary electron multiplier. Errors in the isotopic ratios were propagated numerically (Ludwig 1980) and are quoted at the 2σ confidence level in Table 1. All ages were calculated using the isotope abundance ratio and decay constants of Jaffey *et al.* (1971), and common-Pb corrections were made by using a model-Pb composition (Stacey & Kramers 1975) at the determined age of the rock. In all cases, the uncertainty in the amount and composition of the common-Pb calculated in this manner represents an insignificant contribution to the error in the age calculation.

Results

Table 1 lists the U-Pb isotopic data obtained from zircon in seven key rocks from southern Britain, including the first such data from the Warren House Formation (sample 4), the Padarn Tuff Formation (sample 2), and an intrusive granophyric diorite ('markfieldite'; sample 5) at Nuneaton. All analyzed zircons from the dated volcanic and sub-volcanic rocks (samples 2, 4, 6 and 7) are free of an inherited zircon component (or contain only minute amounts) and, because their $^{207}Pb/^{206}Pb$ ages were precisely replicated, they are considered to be reliably dated. For these samples, the cited uncertainty of the age is the 2σ error of the mean $^{207}Pb/^{206}Pb$ age. The intrusive igneous rocks of the Coedana Granite (sample 1) and Malvernian Complex (sample 3), however, contained variable amounts of older, inherited zircon, possibly as whole xenocrysts or invisible cores in any single grain, and their ages are somewhat less precise. For these rocks, we cite concordia intercept ages and errors calculated with the algorithm of Davis (1982).

Coedana Granite, Anglesey. Sample 1 is a coarse-grained porphyritic variety of the Coedana Granite collected from the quarry at Gwalchmai where it intrudes high-grade pelitic gneiss and amphibolite of the central zone of the Mona Complex (Greenly 1919; Shackleton 1969, 1975). Some rocks of the Mona Complex underwent early blueschist-

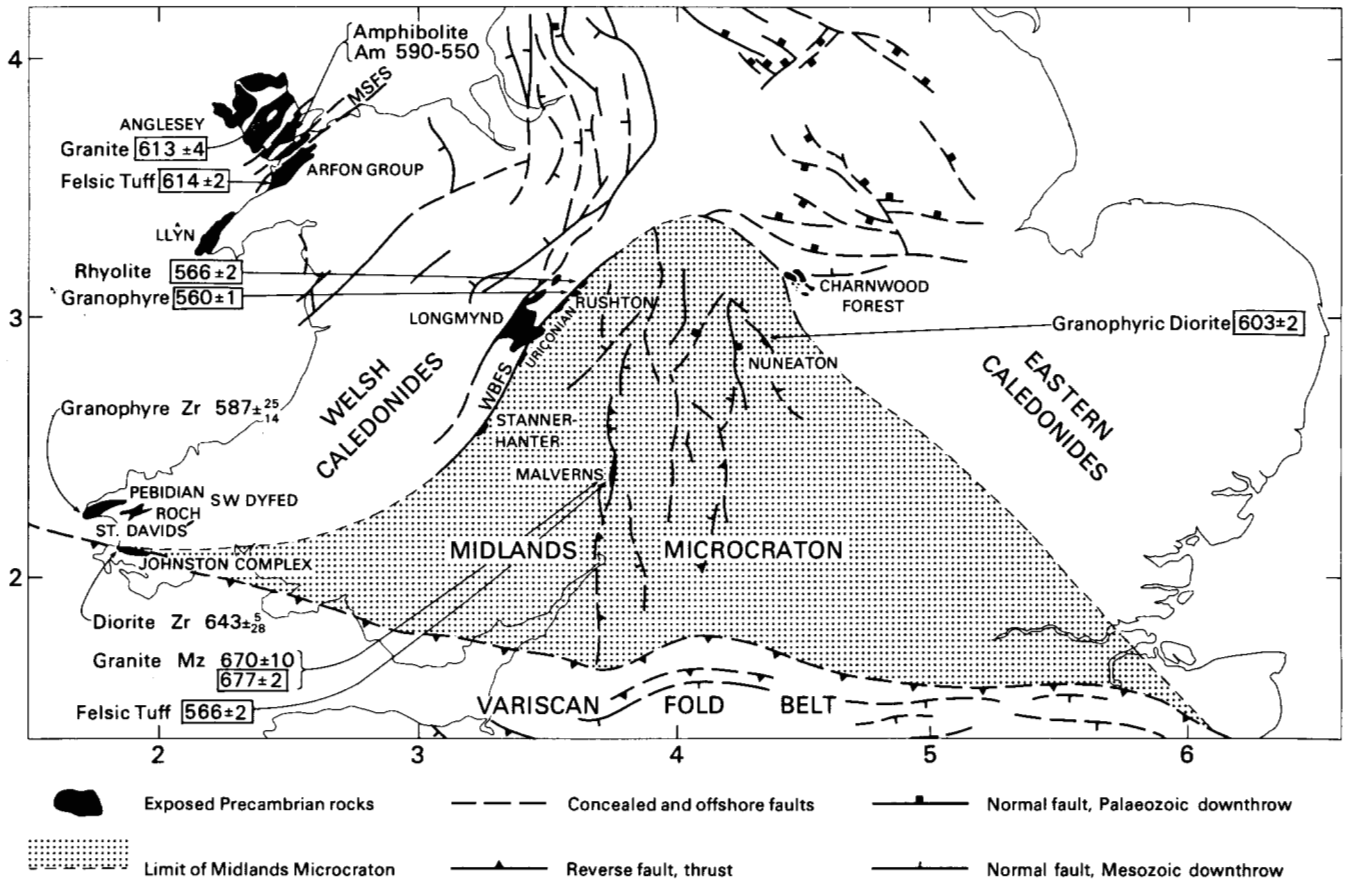


Fig. 1. Sketch map of southern Britain showing the location of exposed Precambrian rocks and major (Palaeozoic–Mesozoic) tectonic lineaments, after Smith (1985) and Lee *et al.* (1990). The suites investigated here are indicated (with the new data shown in boxes), together with those from which precise ages have been previously reported. Zr and Mz, U-Pb zircon (Patchett & Jocelyn 1979) and monazite (Thorpe *et al.* 1984) ages; Am, $^{40}Ar/^{39}Ar$ amphibole ages (Dallmeyer & Gibbons 1987). MSFS, Menai Straits Fault system; WBFS, Welsh Borderland Fault System.

facies metamorphism (dated at *c.* 560–550 Ma, Dallmeyer & Gibbons 1987), but contact relationships of the Coedana Granite are obscure (Shackleton 1969), and it is unclear whether blueschist-facies metamorphism preceded or followed granite emplacement, dated at 603 ± 34 Ma by the Rb-Sr whole-rock isochron method (Beckinsale & Thorpe 1979). The Coedana Granite appears to postdate much of the deformation in the host rocks, but it is affected locally by a ductile high-strain zone along its southern margin (Mann 1986).

The zircons from sample 1 consist of clear, flat prisms (length/breadth ≈ 4) and short-prismatic faceted grains, approximately 100–75 μm long. Analyses of five slightly paramagnetic zircon fractions from both populations (1a–e, Table 1) define a discordia with intercept ages of 613 ± 4 Ma and 1443 +37/−34 Ma (Fig. 2), interpreted as the igneous crystallization age and the mean age of an inherited zircon component, respectively.

Padarn Tuff Formation (*Arfon Group*). Sample 2 is a homogeneous and massive rhyolitic ash-flow tuff within the Padarn Tuff Formation, the lowest stratigraphical formation of the Arfon Group (Reedman *et al.* 1984), collected at the outlet of Llyn Padarn near the village of Bryn'refail. Higher rocks in the Arfon Group are largely epiclastic, containing clasts of Monian metasedimentary rocks and ash-flow tuff from the Padarn Tuff Formation, and they are overlain conformably by the Llanberis Slates Formation. Near the top of the Llanberis Slates Formation, trilobites including *Pseudatops viola* and other forms (Howell & Stubblefield 1950; Wood 1969) have been found, which have been correlated with the top of the 'Olenellid' zone in the middle of the Lower Cambrian Comley Series (Cowie *et al.* 1972; Bassett *et al.* 1976). On these grounds the Padarn Tuff Formation has been assigned a late Precambrian or possibly earliest Cambrian age (Reedman *et al.* 1984; Gibbons 1987).

The zircons in sample 2 consist of clear, multi-faceted and generally small (<100 μm) short-prismatic grains. Because the zircon yield from the *c.* 12 kg sample was fairly low, only small (<40 μg) multi-grain fractions were analysed. Three such fractions (2a–c, Table 1) from the slightly paramagnetic populations are concordant at 614 ± 2 Ma (Fig. 3), which is interpreted as the eruption age of the rhyolite ash-flow tuff. The replicated concordant analyses further indicate that the analysed grains are free of an inherited zircon component, and did not experience detectable isotopic disturbance.

Malvernian complex, Malvern Hills. The Malvernian Complex is a heterogeneous rock suite consisting of diorite, granodiorite, granite and hybridized equivalents of general

Table 1. *U-Pb zircon data, Precambrian rocks of southern Britain*

Fractions		Concentrations				Atomic ratios							Age [Ma]	
Properties No.	(1)	Wt. [mg]	Pb rad [ppm] (2)	U [ppm] (2)	Pb common [pg] (3)	$^{206}Pb/^{204}Pb$ (4)	$^{207}Pb/^{206}Pb$ (5)	$^{208}Pb/^{206}Pb$ (5)	$^{206}Pb/^{238}U$ (6)		$^{207}Pb/^{235}U$ (6)		$^{207}Pb/^{206}Pb$ (6)	
North Wales														
1. RT-89A-22 [Coedana granite, Gwalchmai Quarry, Anglesey] [SH 381 768][7]														
a	M@0 – 100 + 150,cl,e,sp,A	0.145	47.4	379.2	9	45,796	0.06946	0.1351	0.12068	±0.55	1.1491	±0.56	900.5	±3.0
b	M@0 – 100 + 150,cl,fp,A	0.022	33.7	294.4	13	3,334	0.06597	0.1945	0.10970	±0.24	0.9799	±0.25	767.4	±2.3
c	M@1 – 100 + 200,cl,fp,A	0.035	47.2	402.6	25	3,799	0.06737	0.1976	0.10859	±0.48	0.9666	±0.56	759.9	±3.3
d	M@1 – 150 + 200,cl,fp,A	0.172	41.9	380.3	17	24,053	0.06278	0.1778	0.10342	±0.56	0.8868	±0.57	680.6	±3.1
e	M@1 – 150 + 200,cl,fp,A	0.162	45.2	405.9	12	36,415	0.06239	0.1884	0.10352	±0.47	0.8844	±0.48	672.8	±3.0
2. RT-89NW-21 [Padarn Tuff Formation, Arfon Group, Bryn'refail, Wales] [SH 559 626][7]														
a	M@1 – 150 + 200,cl,sp,A	0.011	20.8	155.1	4	2,901	0.06269	0.5000	0.09958	±0.38	0.8287	±0.44	616.3	±3.8
b	M@1 – 150 + 200,cl,sp,A	0.015	19.5	148.5	4	3,652	0.06250	0.4635	0.09976	±0.38	0.8289	±0.46	613.0	±4.9
c	M@3 – 100 + 150,cl,sp,A	0.040	20.7	162.1	7	5,713	0.06115	0.4316	0.09954	±0.38	0.8273	±0.44	613.8	±3.2
Malvern Hills, England														
3. RT-89MH-27 [Deformed Granite, Malvernian complex, North Malvern] [SO 7691 4701][7]														
a	NM@0 – 150 + 200,cl,lp,A	0.025	20.5	160.4	15	1,815	0.06527	0.2872	0.11053	±0.24	0.9444	±0.29	675.1	±3.0
b	NM@0 – 150 + 200,cl,lp,A	0.098	24.3	191.0	66	1,989	0.06869	0.2797	0.11121	±0.28	0.9563	±0.30	686.7	±3.0
c	M@0 – 150 + 200,cl,lp,A	0.113	23.7	185.8	15	9,305	0.06327	0.2626	0.11193	±0.25	0.9678	±0.27	698.5	±2.0
d	M@0 – 150 + 200,s,sp,A	0.060	22.0	153.5	4	20,855	0.07164	0.2100	0.13041	±0.44	1.2797	±0.50	962.3	±3.1
e	M@0 – 150 + 200,cr,lp,A	0.068	18.8	150.8	5	13,617	0.06392	0.2567	0.11002	±0.46	0.9608	±0.50	719.7	±3.2
4. RT-89MH-26 [Felsic Tuff, Warren House Formation, Broad Down] [SO 7655 3960][7]														
a	M@0 – 150 + 200,cl,sp,A	0.029	10.1	99.3	3	5,179	0.05702	0.2324	0.09190	±0.44	0.7471	±0.51	565.7	±4.1
b	M@0 – 150 – 200,cl,sp,A	0.012	10.9	107.4	2	3,269	0.05825	0.2330	0.09191	±0.46	0.7478	±0.54	567.6	±2.4
c	M@0 – 150 + 200,cl,sp,A	0.059	10.9	106.4	13	2,799	0.06024	0.2349	0.09199	±0.52	0.7474	±0.58	564.5	±4.0
Nuneaton, England														
5. TCP-89-6 [Granophyric diorite 'Markfieldite'] [SK 3422 9322][7]														
a	M@0 – 150 + 200,cl,lp,A	0.010	13.8	120.1	3	2,208	0.05825	0.2917	0.09834	±0.44	0.8132	±0.52	602.6	±3.4
b	NM@0 – 150 + 200,cl,b,e,A	0.003	21.9	195.4	2	1,710	0.05882	0.2549	0.09827	±0.48	0.8131	±0.56	603.9	±3.6
Shropshire, England														
6. RT-89SH-24 [Rhyolite lava, Uriconian volcanic group, Overley Hill] [SJ 6128 1112][7]														
a	M@0 – 150 + 200,cl,sp,A	0.016	14.6	145.8	7	1,823	0.05946	0.2162	0.09144	±0.40	0.7435	±0.48	566.1	±4.5
b	M@1 – 150 + 200,cl,sp,A	0.039	13.4	140.3	9	3,328	0.05814	0.1581	0.09116	±0.66	0.7404	±0.82	563.7	±5.4
c	M@1 – 150 + 200,cl,sp,A	0.040	13.7	134.1	12	2,671	0.05910	0.2411	0.09136	±0.40	0.7432	±0.46	567.0	±4.0
7. RT-89SH-23 [Ercall granophyre, Uriconian volcanic group, Wrekin Quarry] [SJ 6442 0970][7]														
a	M@0 – 200 + 325,cl,e,pb,A	0.009	23.7	222.3	9	1,328	0.06369	0.3121	0.09029	±0.44	0.7321	±0.49	559.9	±3.2
b	M@1 – 200 + 325,cl,sp,pb,A	0.024	22.8	207.5	7	3,903	0.05912	0.3522	0.09067	±0.44	0.7353	±0.52	560.4	±3.5
c	M@1 – 200 + 325,cl,sp,pb,A	0.014	23.9	216.2	5	3,242	0.05907	0.3544	0.09092	±0.30	0.7371	±0.34	559.6	±2.9
d	M@1 – 200 + 325,cl,sp,pb,A	0.010	21.6	273.7	4	2,560	0.06122	0.3605	0.09093	±0.19	0.7377	±0.32	561.2	±4.5
e	M@1 – 200 + 325,cl,sp,A	0.036	24.4	226.9	13	3,471	0.06084	0.3120	0.09105	±0.44	0.7422	±0.52	571.4	±4.0
f	M@0 – 200 + 325,cl,sp,A	0.009	27.2	250.1	4	3,012	0.06111	0.3111	0.09255	±0.44	0.7574	±0.52	580.2	±3.4

Notes:

[1] NM@0, M@0, paramagnetic at stated degree of tilt at full magnetic field on Frantz Isodynamic Separator; 100, 200, size in mesh; b, broken; cl, clear; cr, cracked; e, euhedral; fp, faceted prismatic; lp, long-prismatic; pb, pale brown; s, subrounded; sp, short-prismatic; A, air abraded following Krogh (1982).

[2] concentrations are known to ±5% for sample weights of about 0.1 mg and ±10% for samples ≤ 0.05 mg.

[3] corrected for 0.0215 mole fraction common Pb in the ^{205}Pb spike.

[4] measured ratio corrected for fractionation only; Pb and U fractionation correction = 0.1%/amu (±0.05%).

[5] corrected for fractionation and blank; U blank = 1 pg; Pb blank ≤ 4 pg.

[6] corrected for fractionation, spike, blank, and initial common Pb. 2-sigma errors in the Pb/U ratios (%) and $^{207}Pb/^{206}Pb$ ages (absolute) are calculated following Ludwig (1980).

[7] Universal Transverse Mercator (UTM) grid coordinates on 1:50 000 Ordnance Survey Maps.

calc-alkaline affinity (Lambert & Holland 1971; Thorpe *et al.* 1984). Published isotopic ages of the Malvernian Complex include a U-Pb monazite age of 670 ± 10 Ma (Thorpe *et al.* 1984) and a Rb-Sr whole-rock isochron age of 681 ± 53 Ma (Beckinsale *et al.* 1981), both interpreted as the time of igneous emplacement. The sample dated in this study (sample 3) is a medium-grained deformed granite from the North Malvern Quarry, collected where the complex is strongly deformed by localized ductile shear and brittle deformation zones.

Zircons in sample 3 consist of generally clear subhedral and euhedral crystals, including subrounded short-prismatic and long-prismatic grains. One fraction of short-prismatic, faceted grains (3d) and four fractions of elongate long-prismatic grains (3a–c and 3e) were selected for analysis.

One analysis of a fraction of long-prismatic zircon is concordant at 675 Ma (3a), and two others are less than 3% discordant (3b–c). All three are colinear with an analysis of short-prismatic grains (3d), defining a discordia with intercept ages of 677 ± 2 Ma and 1598 +32/−30 Ma (Fig. 2). A fifth fraction of long-prismatic grains (3e), does not fit this line, possibly because of slight secondary Pb-loss. The lower and upper intercept ages are interpreted as the time of granite emplacement and the average age of an inherited Proterozoic zircon component, respectively.

Warren House Formation, Malvern Hills. Sample 4, collected from Broad Down, is a crystal–lithic tuff of rhyolitic composition occurring within a sequence of basaltic and basaltic–andesite lavas, in presumed fault contact with the Malvernian Complex (Worssam *et al.* 1989). No

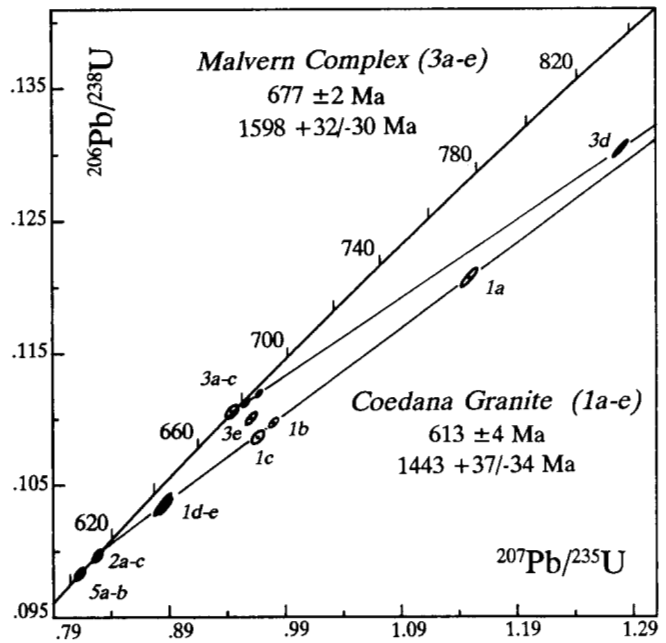


Fig. 2. Concordia diagram of U-Pb zircon analyses of samples 1 (Coedana Granite) and 3 (Malvernian Complex). Data from samples 2 (Padarn Tuff Formation, Arfon Group) and 5 (Granophyric diorite, Nuneaton), shown for reference purposes.

previous isotopic age has been published for the Warren House Formation, but a Precambrian age has been inferred (Earp & Hains 1971) based on pebbles, attributed to the group, within the overlying Malvern Quartzite. The latter and the lower beds of the overlying Hollybush Sandstone

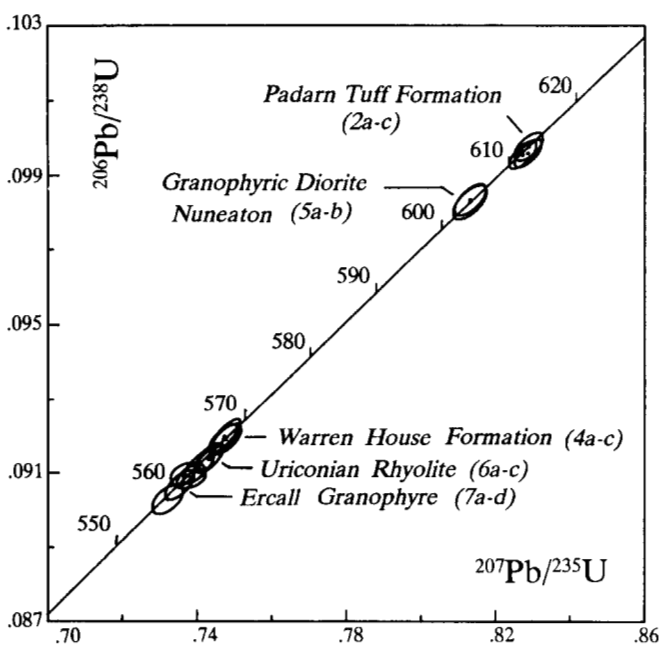


Fig. 3. Concordia diagram of U-Pb zircon analyses of samples from the Arfon Group (2a–c), Nuneaton granophyric diorite (5a–b), Warren House Formation (4a–c), Uriconian volcanic group (6a–c), and the Ercall Granophyre (7a–d) Analyses 7e–f are omitted for reasons of clarity.

contain '*Obolella*' *groomi, Paterina* and hyolithids indicating an age near the Tommotian-Atdabanian boundary, probably of lower Atdabanian age (Cowie *et al.* 1972).

Sample 4 contains a homogeneous zircon population consisting of generally clear, short-prismatic crystals. Three zircon fractions (4a–c), each composed of 6–20 grains, yielded overlapping and concordant analyses defining an eruption age for the tuff of 566 ± 2 Ma (Fig. 3).

Granophyric diorite, Nuneaton. Sample 5 is a granophyric diorite intrusion in the Caldecote Volcanic Formation at Nuneaton. Both units are overlain unconformably (Wills & Shotton 1934) by the *Hyolithes*-bearing Hartshill Quartzite Formation of Early Cambrian (Atdabanian, or possibly latest Tommotian) age (Brasier *et al.* 1978; Brasier & Hewitt 1981; Brasier 1984). The Caldecote Volcanic Formation is correlated with the Charnian Supergroup (Dunning 1975). Sample 5 is petrographically and geochemically similar to the 'markfieldite' or Southern Diorite suite that occurs throughout the Charnwood area as a series of massive, laccolith-like intrusions emplaced prior to development of penetrative cleavage. A minimum age can be assigned to the Charnian Supergroup based on the Rb-Sr whole-rock date of 540 ± 57 Ma (Cribb 1975; recalculated by Thorpe *et al.* 1984 using $^{87}Rb = 1.42 \times 10^{-11}\ a^{-1}$) for the Southern Diorite suite.

Zircon from sample 5 consists of clear euhedral prismatic grains with elongate prismatic forms or of broken euhedral crystals. Analyses of both elongate prisms (5a) and broken euhedral crystals (5b) are concordant at 603 ± 2 Ma (Fig. 3), which is interpreted as the emplacement age of the granophyric diorite, and a minimum age of the Caldecote Volcanic Formation.

Uriconian volcanic group, Shropshire. Sample 6 is a rhyolite lava, exhibiting flowage banding and folding, collected from high in the Wrockwardine succession of the eastern Uriconian volcanic group near Overley Hill. The group consists of lavas and tuffs of basaltic, basaltic-andesite, dacitic and rhyolitic composition (Pharaoh *et al.* 1987) exposed on a number of hills adjacent to the Welsh Borderland Fault System (Woodcock & Gibbons 1988, Fig. 1). Rhyolitic tuffs in this group have yielded a Rb-Sr whole-rock isochron age of 558 ± 16 Ma (Patchett *et al.* 1980), interpreted as an igneous crystallization age, although K-Ar ages as old as 680 Ma have been reported (Fitch *et al.* 1969).

Zircons from sample 6 consist of sharply faceted, short-prismatic grains similar to, but generally smaller than, zircons of sample 4 from the Warren House Formation. Three analyses of well-faceted short-prismatic zircon from nonmagnetic and slightly paramagnetic populations yielded concordant and overlapping ages defining the eruption age of 566 ± 2 Ma (6a–c, Fig. 3), in good agreement with the Rb-Sr whole-rock isochron age. As was the case with the analyses of samples 2, 4 and 5, no detectable amount of zircon inheritance or secondary Pb-loss is indicated by the isotopic data.

Ercall Granophyre, Shropshire. Sample 7 is from the Ercall Granophyre, a small granitic intrusion of possible subvolcanic origin, intrusive into Uriconian felsic volcanic rocks at the Ercall Quarry near the Wrekin in Shropshire. The Wrekin Quartzite (Lower Cambrian) lies

unconformably above the eroded and weathered surface of the granophyre (Cope & Gibbons 1987), and contains trace fossils (*Diplocraterion*, Brasier & Hewitt 1981) of possibly early Atdabanian to late Tommotian age. The Wrekin Quartzite is overlain conformably by the Lower Comley Series of certain early Atdabanian (Cowie *et al.* 1972) and possibly late Tommotian age (Rushton 1972; Bengtson 1977).

The Ercall Granophyre has yielded a Rb-Sr whole-rock isochron age of 533 ± 13 Ma (Patchett *et al.* 1980), which was interpreted as the age of emplacement. This age was confirmed by Thorpe *et al.* (1984) as 534 ± 10 Ma by additional Rb-Sr analyses and further corroborated by Compston *et al.* (1984), who cite an age of 531 ± 5 Ma for the granophyre based on replicated $^{206}Pb/^{238}U$ zircon ages determined using the SHRIMP ion-microprobe. Conventional U-Pb analyses of two multi-grain zircon fractions by Compston *et al.* (1984) yielded reproducible $^{207}Pb/^{206}Pb$ ages of 565 ± 7 Ma, interpreted by these authors as being older than the true emplacement age because of the presence of inherited zircon.

Zircons recovered from the Ercall Granophyre are generally very small (<75 μm), short-prismatic equant grains, commonly coloured pale yellow-brown. Some grains have obvious fluid or melt inclusions, but zircon cores were not apparent in any grains observed at ×100 magnification. Two rather small zircon fractions (7–8 grains, analyses 7a and 7f, Table 1) were analysed initially; analysis 7a is concordant at 560 Ma, whereas 7f contains a minute amount of inherited zircon and plots below concordia with a $^{207}Pb/^{206}Pb$ age of 580 Ma (Table 1). As a check of the reliability of the single concordant analysis, four additional fractions of highest-quality short-prismatic zircon (10–20 grains each) were analysed. Three fractions (7b–d) are concordant at *c.* 560 Ma, whereas the other (7e) contains a trace component of older, inherited zircon with a minimum age of 571 Ma (Table 1). Based on the replicated concordant analyses (7a–d, Fig. 3), emplacement of the Ercall Granophyre occurred at 560 ± 1 Ma, only slightly later than eruption of the Uriconian Volcanic Group at Overley Hill (566 ± 2 Ma) and the Warren House Formation at Broad Down (566 ± 2 Ma).

Discussion

The age of igneous events in North Wales and Anglesey

The ages of 613 ± 4 Ma for the Coedana Granite and 614 ± 2 Ma for the Padarn Tuff Formation of the Arfon Group confirm a late Precambrian age for both units, but throw into doubt the interpretation of the *c.* 560–550 Ma $^{40}Ar/^{39}Ar$ age as the time of blueschist-facies metamorphism of the Penmynydd schist, Anglesey (Dallmeyer & Gibbons 1987). The blueschists were previously correlated with other Monian schists (the Penmynydd Zone of Greenly 1919), but are now interpreted as a remnant of a displaced terrane within the Menai Strait Fault System (Gibbons 1987). The juxtaposition of different tectonic units along this fault system is considered to have occurred prior to *c.* 550 Ma (Gibbons 1987), the cooling age following blueschist metamorphism (Dallmeyer & Gibbons 1987), and the approximate age of emplacement of the Sarn Complex of Llŷn (549 ± 19 Ma, Beckinsale *et al.* 1984). Because fragments of Monian-like schist occur in a conglomerate within the upper part of the Arfon Group, and because Arfon-like rocks (i.e. Bwlch Gwyn 'felsite', Baron Hill and Careg Onen beds, and the Trefdraeth Conglomerate) unconformably overlie the Mona Complex and the Berw Shear Zone (Greenly 1919; Wood 1969; Reedman *et al.* 1984), the inferred late Precambrian age of the Padarn Tuff Formation has been questioned (Gibbons 1987) even though it lies more than 1·6 km stratigraphically beneath the Llanberis Slates Formation (Lower Cambrian).

The new zircon ages for the Padarn Tuff Formation and Coedana Granite indicate that metamorphism of the Mona Complex and terrane juxtaposition are older than 614 ± 2 Ma, a *minimum* age for the *Late Precambrian* deformation and metamorphism on Anglesey. Moreover, these ages reconfirm the late Precambrian age of the Padarn Tuff Formation, only inferred previously on stratigraphical grounds, and also suggest that the Coedana Granite and Padarn Tuff Formation may be coeval.

The age of igneous events in Southern England

The oldest isotopic age reported here is for the deformed granite from the Malvernian Complex, emplaced at 677 ± 2 Ma. This age agrees well with the U-Pb monazite age of 670 ± 10 Ma (Patchett cited in Thorpe *et al.* 1984) as well as the Rb-Sr whole-rock isochron age of 681 ± 53 Ma (Beckinsale *et al.* 1981), thus confirming the generally accepted late Precambrian age for this complex. The mean age of the inherited zircon component in sample 3 is 1598 +32/−30 Ma, and the minimum age is *c.* 962 Ma, the $^{207}Pb/^{206}Pb$ age of the most discordant fraction (3d, Table 1). The source of this component is unknown but it may reflect the presence of older crust, or rocks derived from older crust, underlying southern England. Other members of the oldest calc-alkaline plutonic complexes in southern Britain (e.g. the Johnston Diorite Complex of south Wales) have yielded isotopic ages in the range 700–640 Ma (Patchett & Jocelyn 1979; Thorpe *et al.* 1984); the relationship between these rocks and the markedly younger volcanic suites warrants further study.

Volcanic rocks of the Uriconian volcanic group (Shropshire) and Warren House Formation (Malvern Hills) yield indistinguishable isotopic ages of 566 ± 2 Ma. On the basis of the Ercall granophyre age, this confirms the Precambrian age for these rocks inferred previously by Earp & Hains (1971). Pharaoh *et al.* (1987) considered that differences in the chemical composition of the two volcanic groups reflect their origin in different tectonic environments. The Uriconian volcanic group is a generally bimodal suite exhibiting both within-plate and volcanic-arc trace-element signatures, and it probably formed in an ensialic back-arc marginal basin underlain by juvenile continental lithosphere (as represented by the Rushton Schist). The Warren House Formation lacks a within-plate trace-element signature, having a source more depleted in incompatible elements, and the group is compositionally similar to lavas erupted in modern oceanic marginal basins and Pan-African ophiolitic crust in North Africa and the Middle East (Pharaoh *et al.* 1987). An ophiolitic association has not been demonstrated in either suite. The ages demonstrate that both suites are contemporaneous products of subduction-related magmatism, possibly marginal to the same volcanic arc, in a magmatic cycle post-dating eruption of the Charnian volcanic rocks.

The age of 603 ± 2 Ma for the granophyric diorite ('markfieldite'), which intrudes the Caldecote Volcanic Formation at Nuneaton, dates the final stages of Charnian igneous activity in the eastern part of the Midlands microcraton. This age is more precise and slightly older, at the maximum range of quoted uncertainty, than the Rb-Sr age for the correlative Southern Diorite suite of Charnwood (540 ± 57 Ma, Cribb 1975, *cited in* Thorpe *et al.* 1984).

The Cambrian–Precambrian boundary in southern Britain

The youngest U-Pb zircon age determined is for the Ercall Granophyre (560 ± 1 Ma), which intrudes the Uriconian volcanic group, and lies unconformably beneath the Wrekin Quartzite of late Tommotian or early Atdabanian (Early Cambrian) age. The zircon age is distinctly older than previously published ages of the granophyre, and the replicated concordant analyses indicate that the age is reliable, despite the trace presence of older inherited zircon. The Ercall Granophyre has long been considered the youngest member of the Uriconian igneous suite, and the new age demonstrates that the Uriconian magmatic cycle in the Welsh Borderlands lasted until at least 560 Ma, after which the area was uplifted, eroded, and chemically weathered (Cope & Gibbons 1987) prior to deposition of the Wrekin and Malvern quartzites. The Longmyndian Supergroup, a sequence of shallow marine and fluviatile sediments at least 6.5 km thick (Pauley 1990), is commonly inferred to occupy a stratigraphical position between the Uriconian volcanic group and the lowest Cambrian strata. Until the tuffaceous horizons in the lower part of the supergroup (e.g. the Buxton Rock and Batch Volcanics) are isotopically dated, however, the age of 560 ± 1 Ma is the most reliable maximum age yet determined for the Precambrian–Cambrian boundary in southern Britain.

Comparisons with Avalonian terranes in North America and other Gondwana-fringing Pan-African terranes

In New Brunswick, Nova Scotia, and especially eastern Newfoundland, the currently recognized Late Precambrian igneous rocks comprising the Avalon Zone (or Avalon Composite Terrane) formed during three main episodes of igneous activity, separated in time by regional unconformities. The volcanic rocks are commonly intercalated with deposits of marine and terrestrial sedimentary rocks (O'Brien *et al.* 1990; Murphy *et al.* 1990), and the upper portion of one such sequence (the Conception Group) is richly fossiliferous with an Ediacaran fauna (Anderson & Misra 1968; Williams & King 1979; Anderson & Conway-Morris 1982). Figure 4 graphically illustrates the two younger episodes of igneous activity currently defined in maritime Canada, and compares these with the U-Pb zircon ages reported here.

The earliest episode is represented by the Burin Group in eastern Newfoundland (Strong *et al.* 1978), a *c.* 5 km thick sequence of submarine mafic volcanic and sedimentary rocks and comagmatic gabbro with a stromatolite-bearing carbonate olistostrome horizon near its base. A thick gabbro sill (Wandsworth Gabbro), forming an integral part of the group, has been dated at 762 ± 2 Ma by the U-Pb zircon method (Krogh *et al.* 1988), essentially coeval with *c.* 760 Ma Pan-African ophiolitic rocks (Bou Azzer) of North Africa (Clauer 1976; Leblanc 1981). A second episode of igneous activity in maritime Canada, corresponding to the period from *c.* 635 Ma–590 Ma, is represented by subaerial bimodal mafic and felsic volcanic rocks of the Love Cove and Marystown groups and part of the Harbour Main Group in Newfoundland (Krogh *et al.* 1988; O'Brien *et al.* 1989), the intermediate felsic tuffs of Sequence A in New Brunswick (Bevier & Barr 1990), and certain plutonic rocks

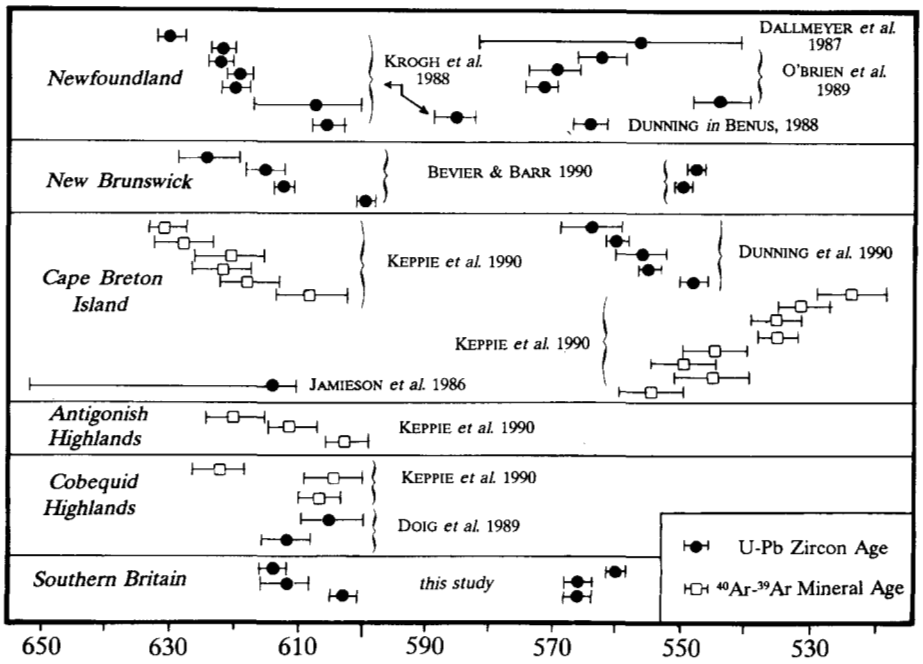


Fig. 4. Compilation of selected isotopic ages for plutonic and volcanic rocks of late Precambrian age in maritime Canada and southern Britain.

from Cape Breton Island (Jamieson *et al.* 1986; Keppie *et al.* 1990) and the Antigonish and Cobequid Highlands of Nova Scotia (Keppie *et al.* 1990; Doig *et al.* 1989) (Fig. 4). Intrusive igneous rocks of general calc-alkaline affinity were emplaced into the volcanic sequences in eastern Newfoundland and New Brunswick over a similar age range. A third episode of igneous activity, commencing at *c.* 572 Ma and lasting until approximately 548 Ma (possibly as young as *c.* 527 Ma), is represented by the Musgravetown (O'Brien *et al.* 1989) and Rock Harbour groups (Krogh *et al.* 1988) as well as the Swift Current Granite (Dallmeyer *et al.* 1981, recalculated by Krogh *et al.* 1988 to an age of $557 +29/-16$ Ma) in the central Avalon Zone of Newfoundland, Sequence B of eastern New Brunswick (Bevier & Barr 1990), plutonic rocks of the Bras d'Or terrane of northern Cape Breton island (Dunning *et al.* 1990) as well as other plutonic rocks in southern Cape Breton Island (Keppie *et al.* 1990) and south Newfoundland (Dunning & O'Brien 1989).

When combined with our new ages from Britain, these data suggest that the Avalonian terranes of Britain and maritime Canada may have shared a common geological history throughout much of late Precambrian time. The *c.* 760 Ma phase of ophiolite development has not yet been recognized in southern Britain, although it is conceivable that possible ophiolitic components of the Mona Complex on Anglesey could be this old. The zircon ages of the Padarn Tuff Formation and the Coedana Granite demonstrate that high-pressure metamorphism of Monian blueschists and their juxtaposition with the Gwna melange occurred prior to 614 ± 2 Ma. There is also evidence of a phase of igneous activity in Britain lasting from about 700–640 Ma, represented by relicts of calc-alkaline plutonic complexes in south Wales (Johnston Diorite complex; $643 +5/-28$ Ma, Patchett & Jocelyn 1979), the Welsh Borderlands (?Stanner–Hanter complex; 702 ± 8 Ma, Patchett *et al.* 1980), and in the central part of the Midlands microcraton (Malvernian complex; 677 ± 2 Ma, Fig. 2). Evidence of zircon inheritance in the Malvernian complex may indicate the existence of significantly older crust underlying parts of southern Britain, not yet dated directly, as is also suggested by the identification of detrital Proterozoic zircons in Caradocian (Longvillian) strata of central Wales (Tucker *et al.* 1990).

Other zircon ages fall in the second episode of igneous activity in maritime Canada, between *c.* 613 and *c.* 603 Ma. In Britain, rocks of this episode include the Coedana Granite, the Padarn Tuff Formation of the Arfon Group, the granophyric diorite ('markfieldite') at Nuneaton, and possibly the Caldecote Volcanic Formation and Charnian Supergroup. A still younger phase of arc magmatism in southern Britain, possibly involving the subduction of oceanic lithosphere from about 566 Ma to 560 Ma, is represented by the Warren House Formation and Uriconian volcanic group, and the Ercall Granophyre. Rocks of this approximate age in eastern Newfoundland include alkaline to peralkaline volcanic rocks found immediately above the sub-Musgravetown angular unconformity (O'Brien *et al.* 1990), and also strongly peralkaline plutonic rocks in the western Avalon zone of Newfoundland. At present, it is premature to make definitive correlations of igneous suites between the Avalonian terranes of Britain and North America. However, the conformity of ages leaves little doubt that these regions were originally juxtaposed along the same destructive margin in late Proterozoic time.

Conclusions

(1) The new U-Pb zircon ages help to constrain the timing of igneous eruptive and intrusive events over the period 677–560 Ma in southern Britain. Previous K-Ar, ^{40}Ar-^{39}Ar, and Rb-Sr data for many of these rocks have large 2σ uncertainties and some are of questionable reliability: their use has obscured the Precambrian history of the region and generated much confusion regarding the absolute age of the base of the Cambrian System (cf. Cowie & Johnson 1985; Odin *et al.* 1985).

(2) The new U-Pb zircon age of the Padarn Tuff Formation (Arfon Group) and Coedana Granite demonstrates that metamorphism of the Mona Complex and terrane juxtaposition in Anglesey must have occurred prior to 614 ± 2 Ma, assuming that the generally accepted correlation of the Bwlch Gwyn 'felsite' with the Padarn Tuff Formation (Reedman *et al.* 1984) is correct. The previously cited ^{40}Ar-^{39}Ar age of post-blueschist metamorphic cooling on Anglesey (*c.* 560–550 Ma, Dallmeyer & Gibbons 1987) is apparently too young, perhaps because it was obtained from minerals (barrosite/crossite) with very low K_2O abundances and poor Ar-retentivity. The zircon age of the Coedana Granite (613 ± 4 Ma) may suggest a genetic relationship between this intrusion and the volcanic rocks of the Arfon Group (and possibly the Bwlch Gwyn 'felsite'), although this requires further testing.

(3) The new U-Pb zircon age of the Ercall Granophyre (560 ± 1 Ma) is a reliable maximum age estimate for the Early Cambrian (Atdabanian and possibly late Tommotian) strata in central England. This determination agrees well with the preliminary results of U-Pb investigations currently underway in eastern Newfoundland (Mistaken Point Formation of the Conception Group) that have directly dated Ediacaran fauna (e.g. *Charniodiscus concentricus* and *Charnia masoni*) to an age of 565 ± 3 Ma (Dunning *in* Benus 1988). Moreover, the new age of the Ercall Granophyre suggests that it is coeval, or nearly so, with nearby members of the Uriconian volcanic group, as suggested previously on geological grounds.

(4) Agreement between the zircon ages of the Warren House Formation and the Uriconian volcanic group demonstrates that these two groups are broadly coeval, although their distinctive geochemical signatures suggest that they formed in different tectonic settings. The 603 ± 2 Ma age of the granophyric diorite ('markfieldite') clearly shows that the Caldecote Volcanic Formation (and by correlation, the Charnian Supergroup) are resolvably older than the Uriconian volcanic group and Warren House Formation and that both the Caldecote Volcanic Formation and the Charnian Supergroup could be younger than the intrusive igneous rocks of the Malvernian Complex. The ages thus suggest that the Precambrian rocks of Britain formed during several distinct periods of igneous activity. Whether or not these periods represent parts of a prolonged cycle of magmatic activity, or rather episodic cycles separated by periods of quiescence, awaits further study.

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